

Preclass Learning: Effective Visualization (ICA 17)

1. Principles of Visual Communication

- **Clarity and Truthfulness:** The graphic must be instantly understandable and avoid all forms of visual distortion (e.g., proper axis scales, accurate geometric representation).
- **Preattentive Attributes:** Leveraging visual properties (color, size, position, motion) that the viewer processes subconsciously to guide their attention to the most important data point.
- **Maximizing Signal:** Eliminating chart clutter like unnecessary grid lines, excessive labels, and distracting backgrounds.

2. Visual Narrative and Annotation

- **"Title as Takeaway":** Titles should state the conclusion of the chart, not just describe its contents (e.g., "Q3 Sales Dropped 15%" instead of "Quarterly Sales").
- **Annotation:** Using text, arrows, or highlighted regions directly on the chart to draw the viewer's eye to a specific finding, outlier, or comparison.
- **Context:** Clearly label all axes, provide units, and cite the data source to ensure the audience can independently evaluate the information.

3. Audience and Medium

- **Audience Tailoring:** Adjust the complexity and vocabulary of the visualization to match the data literacy level of the audience.
- **Presentation Flow:** Visualizations should be introduced logically and support a clear narrative arc in the presentation.